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Comprehensive Digital Rock Study for Heterogeneous Ratawi Limestone Reservoir in Al-Khafji Field, Saudi Arabia

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Abstract

This study aims to utilize the recent technology of digital rock analysis and compare the results for a heterogeneous carbonate reservoir in Al-Khafji field with physical lab SCAL data from over 50 years ago. It evaluates oil-water relative permeability of sister samples, including end points, and their correlation with water saturation. The study also aims to validate the digital approach, especially for core plugs that could not undergo physical experiments due to various limitations.

Digital rock analysis integrates imaging and computational methods for pore-scale modeling. Initial scans with CT assess rock density and homogeneity, creating a preliminary model. Smaller plugs are then analyzed via Micro-CT for detailed data. SEM with X-ray mineralogy measurements completes the digital rock model, mapping capillary spaces. Digital rock models together with fluid data from PVT reports were used with Density-Functional Hydrodynamics simulator to model fluid flow at the pore scale.

This study uses sister core plugs to the original lab SCAL samples with the focus on closing the gaps in the past core analysis program, validate the new technology and encompass all significant flow units while honoring exact reservoir conditions. Digital rock modeling was first performed using the same protocol as per the past lab study. A digital study was conducted in a blind manner, with the experimental data being shared only after the modeling results were achieved on five plugs of different permeability ranges.

Relative permeability data acquisition was the focus of the study. The resulting oil-water relative permeability data and their correlation with water saturation from sister samples were analyzed and compared between digital and physical methods. Initial saturations, relative permeability curves and end points were also compared. Good agreement between digital and physical SCAL was observed on most of the samples studied. The obtained digital data was found to be within physical laboratory measurement accuracy limits for the end points, and relative permeability cross-over point saturations.In addition, obtained modeling data was verified using lab MICP and RCA data to confirm rock type classification and grouping and QC constructed digital rock models with routine core analysis measurements of porosity/ permeability and pore throat distribution data.

As the model was validated by comparing obtained digital SCAL data to the most representative plugs from the 50 year old study, it also allowed to extend the characterizations to samples that could not be measured in the previous SCAL study due to limitations of lab equipment in that era.

Considering the challenges in updating and modernizing the old legacy data for effective utilization in reservoir and fluid modeling, digital rock modeling offers a faster, more cost-effective solution that provides updated information to older core samples or those lacking live PVT data. Digital core analysis can help to fill the gaps and enable fast decision making in field development, enhanced oil recovery and early exploration projects. It provides a timely response to urgent projects while improving recovery strategies.

Introduction

The characterization of carbonate reservoirs remains one of the most complex challenges in petroleum engineering due to their inherent heterogeneity, variable wettability, vertical and/or lateral fluid composition change and intricate pore structures. Traditional Special Core Analysis (SCAL) has long been the cornerstone of reservoir evaluation, providing critical data for fluid flow modeling, Enhanced Oil Recovery (EOR) planning, and reservoir simulation. However, SCAL is time-consuming, expensive, and often limited by sample integrity and laboratory capabilities. In many cases legacy SCAL data—sometimes collected decades ago— remains the only available source of information for certain reservoirs, despite advances in technology and understanding.

Digital Rock Analysis (DRA) is a modern, non-destructive technique relying on microcomputed tomography (micro-CT) data and advanced pore-scale modeling. DRA combines high-resolution imaging with advanced computational modeling to simulate fluid flow through rock samples at the pore scale. By using tools such as CT, micro-CT, SEM with EDX mineralogy information, and FIB-SEM, DRA reconstructs the internal pore structure of core plugs in 3D. With the input of PVT data and fluid property measurements such as oil density, viscosity, and composition these digital models are then used to calculate key petrophysical properties such as porosity, permeability, and relative permeability. This method then offers a fast, repeatable, and cost-effective alternative to traditional laboratory SCAL methods that can be performed even when traditional methods cannot be applied.

The Ratawi formation in Al-Khafji field consists of a carbonate reservoir located in the Arabian Gulf and presents a unique opportunity. The field's core analysis program was conducted over 50 years ago, produced a dataset that has served as the foundation for the reservoir modeling and simulation. While the original SCAL measurements were considered accurate at the time and captured key metrics such as irreducible water saturation (Swi) and residual oil saturation (Sor), they were obtained using laboratory procedures that have since evolved. Today's best practices in SCAL—such as improved fluid compatibility, refined saturation profiling, data processing methods and routines and enhanced measurement repeatability —highlight limitations in legacy data that were previously undetectable. This study addresses the pressing need to update and validate legacy SCAL data using modern techniques.

The current study is conducted with twofold objectives: The first was in validation of DRA technology for the future use of the methods enabling the organization to take advantage of the timely and cost effective solution while ensuring critical inputs to reservoir models are still as accurate as traditional lab methods. The second was in updating reservoir models and development strategies which requires refreshed datasets that align with current laboratory standards.

The study was designed as a blind comparison, where digital modeling was performed without prior access to historical SCAL results. Five sister core plugs were selected to represent a range of permeability values and flow units. Digital workflows were applied to each plug, generating relative permeability curves, saturation endpoints, and pore throat distributions. These results were then compared to the original SCAL data, focusing on key metrics such as Swi, Sor, and the crossover points of relative permeability curves.

The goal was to assess the accuracy and reliability of Digital SCAL (DSCAL) and to determine whether it could serve as a viable replacement or complement to traditional SCAL in reservoir modeling.

In summary, this study uses closely matching core plugs with the focus on closing the gaps in the past core analysis program, proof of concept of the new technology and covering all important flow units while honoring exact reservoir conditions.

Theory and Definitions

The application of DRA and Digital SCAL in field development planning and in reservoir engineering domain relies on the integration of advanced imaging technologies and modeling to simulate fluid flow behavior at the pore scale. This section outlines the theoretical framework and key definitions relevant to the study, providing context for the methodologies and interpretations that follow.

Digital SCAL refers to the use of pore scale hydrodynamical modeling in digital twins of the real rocks that replicates the outcomes of traditional SCAL analysis via simulations on rock digital twins. The process begins with high-resolution imaging of core samples using techniques such as:

- Single and Dual-Energy CT Scanning: Used to assess bulk density, mineralogical contrasts, and internal heterogeneity.
- Micro-CT Scanning: Provides 3D volumetric images at micron-scale resolution, capturing pore geometry and connectivity.
- Scanning Electron Microscopy (SEM) imaging: Offers nanometer-scale imaging for ultra-fine pore systems, particularly in shales and tight and heterogeneous carbonates.
- Energy-Dispersive X-ray Spectroscopy (EDX): Used in conjunction with SEM imaging to map mineral composition and support wettability modeling.
- Advanced Mineral Identification and Characterization System (AMICS): A method developed to enable high speed mineral mapping based on EDX data.

In Al-Khafji field samples all of the above are utilized in developing the final digital rock model. After imaging, Density Functional Hydrodynamics (DFH) (the modeling technique used in Digital SCAL workflow) is applied to model fluid flow directly through the digitally reconstructed pore space of a rock sample. The workflow included generating digital representations of reservoir and injection fluids (digital fluid models) and performing pore-scale simulations of multiphase flow using the DFH Method via the Direct Hydrodynamic (DHD) simulator. The DFH method employs a diffuse interface approximation at fluid/fluid contacts and utilize Helmholtz energy-based thermodynamic model. DFH uses classical mass, momentum, and energy conservation laws with constitutive relations ensuring consistency between hydrodynamic and thermodynamic descriptions of multiphase compositional systems in the frame of the density functional approach, which allows complex flow processes to be directly modelled at the pore scale. The specific expression for the density functional uses square gradients of molar densities, which enables description of surface tension. The thermodynamic state of the mixture is described by means of bulk and surface Helmholtz energies. Where surface Helmholtz energy allow to account for liquid-solid interaction, i.e., wettability and adsorption. The primary variables are molar densities of chemical components, mass velocity of the medium, and internal energy density.

DFH solves fluid flow across the actual 3D pore structures obtained from high-resolution imaging. This approach allows for more accurate prediction of multiphase flow behavior (including oil-water relative permeability) under reservoir-specific conditions. DFH does so using digital representations of reservoir and injection fluids that accounts for fluid phase properties such as viscosity, density, and fluid-fluid interaction data interfacial tension. Digital rock and fluid models are used in the direct hydrodynamic pore-scale flow simulator, DHD. The DHD is a multiphase compositional hydrodynamic simulator that is based on the DFH. DHD has the ability to handle complex fluids, including surfactant and polymer solutions, complex compositional fluids with phase transitions. DHD simulator is the general-purpose pore-scale simulator that enables detailed routine core analysis (RCA), special core analysis (SCAL), and enhanced oil recovery (EOR) agent optimization.

Relative permeability is a dimensionless measure of the effective permeability of a fluid phase in the presence of one or more immiscible fluids. It is a function of fluid saturation and is critical for modeling multiphase flow in porous media. In oil-water systems, the relative permeability curves describe how easily oil and water can flow through the rock as their saturations change during drainage or imbibition processes. Key data points derived from relative permeability tests include:

- Irreducible Water Saturation (Swi): The water saturation at which water remains nearly immobile.
- Residual Oil Saturation (Sor): The saturation at which oil becomes nearly immobile and further oil production is limited or non-existent.
- Oil Relative Permeability at Swi (kro @ Swi): The effective permeability to oil at irreducible water saturation.
- Water Relative Permeability at Sor (krw @ Sor): The effective permeability to water at residual oil saturation.
- Crossover Point: The saturation at which oil and water relative phase permeabilities are equal

Capillary pressure is the pressure difference across the interface of two immiscible fluids in a porous medium. It is a function of pore size, interfacial tension, and wettability. Capillary pressure curves are essential for understanding fluid distribution and displacement efficiency in the reservoir. Pore throat distribution, often derived from Mercury Injection Capillary Pressure (MICP) tests in physical SCAL, provides insight into the size and connectivity of pore constrictions. In DSCAL, equivalent distributions are extracted from the digital rock models. MICP data is used to classify flow units, assess reservoir quality, and in this study was also used to validate the representativeness of the digital models before relative permeability work.

Current Study

One of the central motivations for this study is the recognition that this study was then able to history match the data using legacy methodology via USS method to validate against the old data set and perform modeling at representative reservoir conditions using Steady-State (SS) method and reservoir fluids for updated curves.

During imaging and sample preparation phase, 5 core plugs from the Al-Khafji field were first screened using CT scanning to assess internal heterogeneity. This step ensured that the selected samples were representative of the reservoir's flow units and suitable for high-resolution imaging. Samples were cleaned and routine core analysis was performed to measure plugs helium filled porosity and gas permeability. Plug imaging data allowed to evaluate rock heterogeneity. For many samples, it was observed that heterogeneity existed even at the plug scale and multiple cylindrical mini-plugs of 4-8 mm mini-plugs were taken to represent the various rock textures and porosity-permeability zones. The exemplary schematic portraying the subsampling process is shown in (Fig. 1).

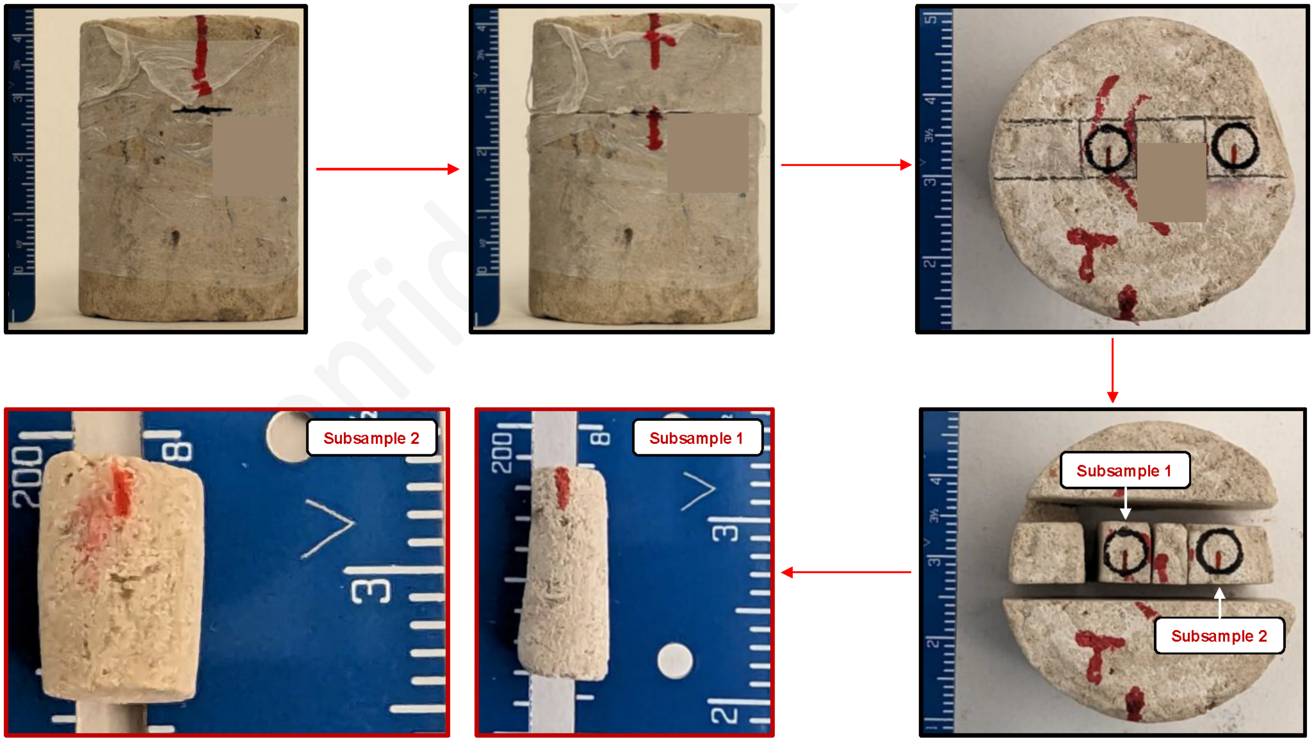

Figure 1—Sample preparation example

In parallel, the remaining sample trim material from the same plugs was sent for MICP analysis. Following initial screening and plugging, micro-CT scanning was performed on sampled mini-plugs providing 3D volumetric images at voxel resolutions ranging from 1.5 to 1 micron per voxel. High-resolution microCT plug scanning was performed with a micro-CT scanner. The scan data was then used to visualize core grains and microCT resolved porosity. High-resolution X-ray microCT scanning provided micron-scale resolution that permitted sample structure visualization. These scans captured the pore geometry and connectivity essential for fluid flow modeling.

SEM imaging was used to resolve ultra-fine pore systems. During the SEM imaging the mini-plug was cut through the middle along the main axis of the mini-plug. SEM samples were mounted on an aluminum SEM stub as a polished round that was milled for more than 8 hours in a Fischione Model 1060 SEM Argon Ion Mill to ensure a horizontal, flat, and polished surface optimal for high magnification SEM imaging. The polished round was sputter-coated with platinum/palladium to reduce charging artefacts and then placed in a Thermo Fisher Teneo FESEM. Overview mosaic imaging was performed at the 200 and 50 nanometer resolution and captured in backscattered electron (BSE) mode. Mineralogy data was acquired using the EDX and AMICS methodology. AMICS software was used to automate mineral identification and phase mapping from EDX data, enabling rapid classification of mineral phases and supporting modeling. AMICS mineralogy results provide information on sample matrix composition, micro-textures, clay minerals, detrital grains, and diagenetic minerals.

The digital rock models of all samples studied were built from shadow projections of X-ray micro-CT and corresponding high-resolution SEM data. The method includes 2D cross-sections reconstruction for each mini plug scan with voxel size equivalent to the micro-CT resolution, regularization, segmentation, and conversion to a rock model.

Reservoir-specific fluid properties (including phase composition, viscosity, and density) were sourced from PVT and fluid analysis reports. In addition, pressure and temperature was set to reflect at 3278 psi and 187 deg F reservoir conditions, fluid composition, sample mineralogy suggested mixed wettability system that was modeled using contact angle range of 120-140 degrees. These inputs ensured that the simulations reflected actual reservoir conditions, enabling accurate prediction of oil-water relative permeability curves and saturation endpoints. The reservoir fluid data selected for this study was taken from a PVT report of a bottomhole sample from Ratawi reservoir in AlKhafji field in an offset well from where the 2023 core was drilled that was taken during formation testing by wireline. The fluid is a black oil sample with API of 31, GOR of 450, and saturation pressure of 1395 psia. The density at reservoir conditions is 0.765 g/cc and viscosity at reservoir conditions is 1.1995 cP. When comparing the fluid to other samples in Ratawi field from various offset wells, the fluid is typical of the black oils in Ratawi and therefore considered a representative input for this study for the oil properties.

The digital rock and digital fluid models were used in Direct Hydrodynamics (DHD) simulator to model fluid flow in sample pores. The performed pore scale modeling allowed to obtain relative permeability curves and analyze saturation dynamics.

After the first sample was analyzed in a blind study the lab protocols of the previous SCAL study was reviewed and it was realized that the legacy lab data was procedurally different from that of the current traditional SCAL studies. Further investigation determined that pre-JBN methods were used to process the unsteady-state flooding data in the report and a large number of tests were performed with refined mineral oil that does not match reservoir oil properties and was used to induce early water breakthrough to extend the saturation range over which relative permeability characteristics could be determined. As a result, the digital modeling protocol was adjusted to perform study in the same manner as in the 1968 report utilizing same fluids, data processing methods and Unsteady-State (USS) flooding instead of Steady-State (SS) one. This allows for better matching of data to historical methods and confirm DRA technology is adequate. However, as discussed previously, the new methodologies provide better overall data quality over wider saturation range, so performed digital study provide the most up to date data to allow updating of the reservoir models with current technology.

To validate the digital models, three stages of validation were undertaken. The first stage is for the data to be compared against routine core analysis data for gas permeability and porosity. Secondly, pore throat distributions derived from digital models were compared with lab MICP data to confirm representativeness. These two stages provide initial confidence in the model and are measured both physically and digitally on the same sample by a commercial lab and in digital rock analysis. Finally, validation included comparing historical SCAL data from the 1968 study. This included matching Swi, Sor, and the crossover points of relative permeability curves. It is important to note that data comparison is from offsetting wells and report is from 1968 whereas the plug samples for DSCAL were taken from a well drilled in 2023. Based on the newly acquired RCA data, matching samples were selected from the 1968 report but some deviation can be expected and considered acceptable. For many samples, it was observed that heterogeneity existed even at the plug scale and multiple cylindrical mini-plugs of 4-8 mm size were taken to represent the various rock textures and porosity-permeability zones.

Experimental Results

Samples were imaged under CT, micro-CT, and SEM as per the process described above. An example (Sample A) of a micro-CT greyscale and segmented volumes and digital rock model ready for Digital SCAL workflow is represented in the Fig. 2.

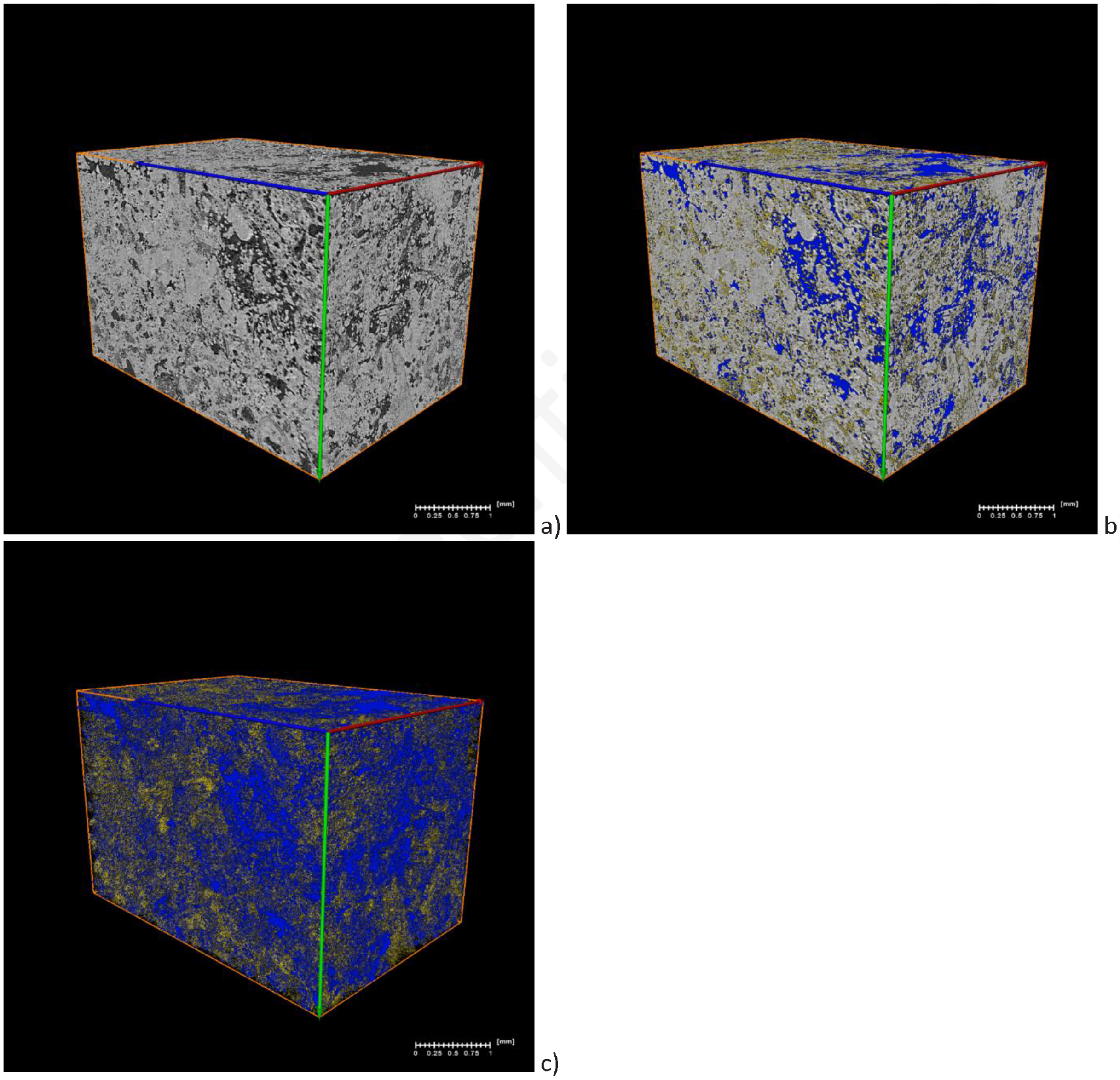

Figure 2—Sample A MicroCT imaged volume (a), segmented volume with blue (micro-CT resolved porosity) and yellow (submicron porosity) (b), constructed digital rock model (c)

SEM imaging was performed on all studied samples. Sample A SEM mosaic images at medium resolution (200nm per pixel) and high resolution (50 nm per pixel) are shown in Fig. 3. Medium resolution mosaic image covers the full dimensions of the sampled mini-plug allowing sample texture analysis at representative scale. The SEM data provided an input to submicron porosity identification and quantification for the micro-CT dataset. SEM site location was registered within micro-CT dataset and used to define porosity values in voxels with submicron porosity to finish construction of the digital rock model.

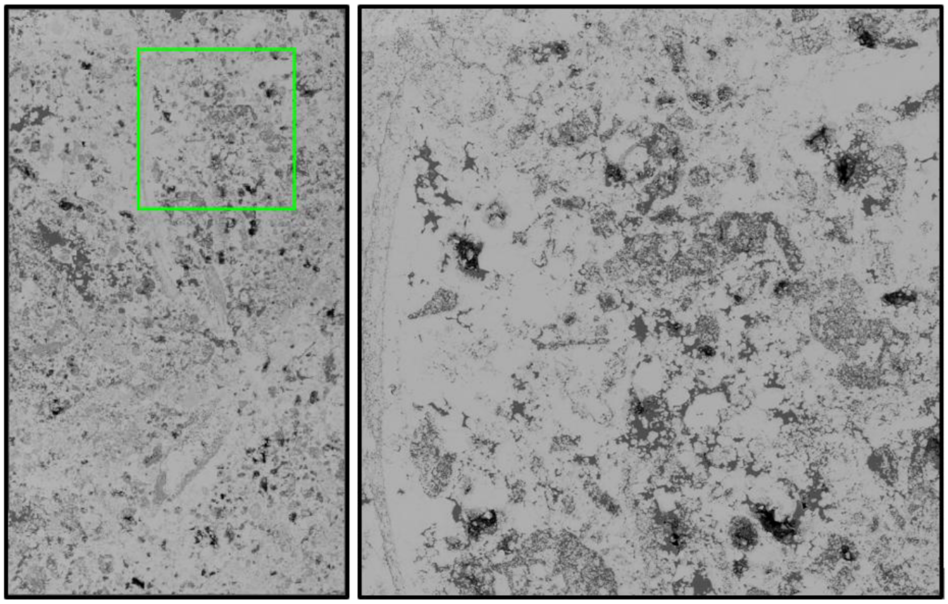

Figure 3—Left to right: medium resolution SEM mosaic imaging (200 nm/pixel) [high resolution imaging location (in green)], high resolution SEM mosaic imaging (50 nm/pixel):

In addition, AMICS mineralogy data was obtained for all studied samples as presented in Fig.4 to confirm that sample mineralogical composition that is highly dominated by calcite at 95% weight with negligible amounts of other minerals (limited dolomite, fe-dolomite and trace amounts of quartz and clays).

Modal Mineralogy EDS AMICS Analysis of Selected Core Samples															
Sample Info	Silicates (Non-Clay)			Carbonates			Accessory Minerals			Clay Minerals		Total			
Sample ID	Quartz	Plagioclase	K-Feldspars	Calcite	Dolomite	Fe-Dolomite / Ankerite	Pyrite	Apatite	Biotite	Illite/Mica	Kaolinite	Total	Total Silicates	Total Carbonates	Total Clays
	wt %	wt %	wt %	wt %	wt %	wt %	wt %	wt %	wt %	wt %	wt %	wt %	wt %	wt %	wt %
A	0.04	0.00	0.00	94.49	0.21	0.19	0.00	0.00	0.00	0.00	0.00	94.94	0.04	94.89	0.00
B-SS1	0.03	0.00	0.00	96.33	0.13	0.07	0.00	0.00	0.00	0.00	0.00	96.56	0.03	96.53	0.00
B-SS2	0.05	0.00	0.00	96.14	0.07	0.03	0.00	0.00	0.00	0.00	0.00	96.30	0.05	96.24	0.00
C-SS1	0.03	0.00	0.00	97.76	0.06	0.02	0.00	0.00	0.00	0.00	0.00	97.88	0.03	97.84	0.00
C-SS2	0.04	0.00	0.00	97.96	0.20	0.11	0.00	0.00	0.00	0.01	0.00	98.32	0.04	98.27	0.01
D-SS1	0.12	0.00	0.00	99.01	0.13	0.09	0.03	0.01	0.00	0.02	0.01	99.43	0.12	99.24	0.03
E-SS1	0.03	0.00	0.00	99.40	0.06	0.00	0.04	0.01	0.00	0.03	0.04	99.60	0.03	99.46	0.07

Figure 4—Summary of mineralogical composition from AMICS mineralogical analysis

The Sample A exemplary digital rock model is shown in Figure 2 (c) with resolved porosity in blue color and submicron porosity in yellow color.

The results of the Digital SCAL workflow were analyzed and compared against both Routine Core Analysis (RCA) and historical SCAL data from the 1968 study. The primary focus was on oil-water relative permeability curves and saturation endpoints. The data presentation is structured to demonstrate the accuracy of the digital models, their alignment with legacy measurements, and their value in updating reservoir models with modern standards.

Before initiating Digital SCAL simulations, each plug underwent RCA to determine helium porosity and gas permeability. These values were used as a baseline to validate the digital rock models derived from imaging. Permeability was determined by simulation of gas flow through the digital rock model using DHD simulator. Across all samples, digital porosity values were near RCA measurements, and gas permeability estimates showed strong correlation. Table 1 and Fig.5 show comparison between lab and digital RCA data. One comment we would like to highlight is that for samples B and C there was a noted heterogeneity in the plug itself, so more than one mini plug was taken. As a result, one can determine that these plugs show the largest difference in values compared to the original RCA plug.

Table 1—Lab and Digital RCA data

Sample ID	Method	Porosity (%)	Permeability (mD)
A	RCA	31.0	24.20
A	Digital	27.7	24.90
B	RCA	26.2	15.80
B	Digital	27.2	13.51
B	Digital	23.7	7.79
C	RCA	21.9	10.70
C	Digital	18.3	12.40
C	Digital	21.8	14.45
D	RCA	21.8	4.71
D	Digital	20.3	4.84
E	RCA	13.0	0.79
E	Digital	13.9	0.59

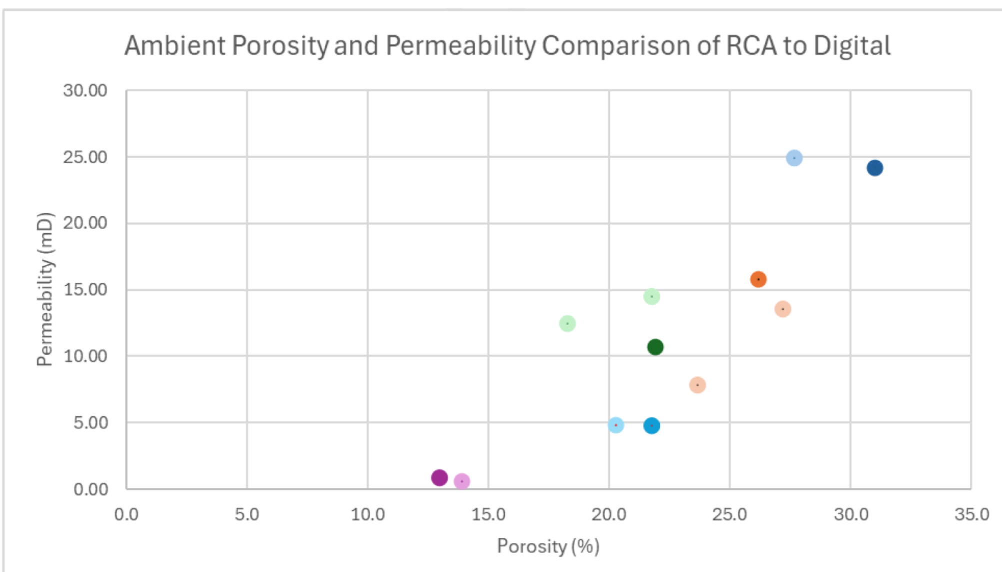

Figure 5—Lab and Digital RCA data (darker shade circle - lab RCA, lighter shade circle - digital RCA)

To assess the representativeness of the digital pore models, digital MICP test was performed, and pore throat distribution was obtained and compared with the lab MICP data on same sample. Mercury-air capillary pressure and pore throat size distribution are shown in Fig. 6. Good match between laboratory and digital MICP data was observed on all studied samples. The digital models successfully captured the dominant pore throat sizes and distribution shapes observed in MICP, confirming that the constructed digital rock models and imaging methods used in study were sufficient for flow simulation. Sample A dataset show an additional local maximum in the MICP pore throat distribution. This is likely due to the sample heterogeneity that was revealed during the plug scanning. It was discovered that selected MICP sample predominantly focuses on the tight subsection of Sample A. Overall we see an excellent match for both pore throat and capillary pressure data.

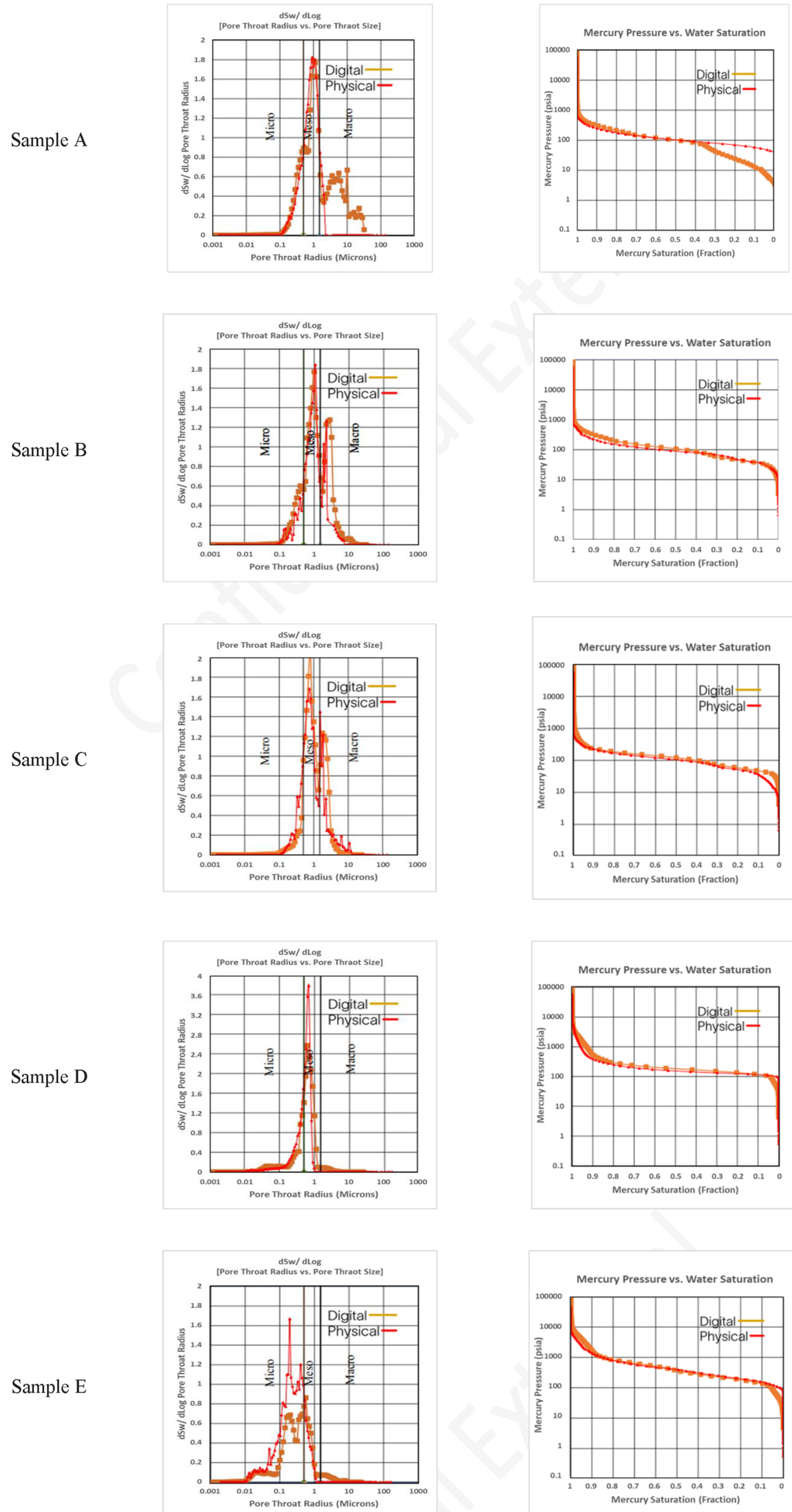

Figure 6—Lab and Digital MICP data. Mercury pore throat radius distribution (left) and pressure versus mercury saturation (right) with curves from digital simulation and physical MICP experiments for all the samples. Legend on the right provide sample IDs.

The relative permeability curves showed a generally good agreement with historical SCAL data (after correction to 1968 methodologies) in terms of shape, crossover points, and endpoint values Fig. 7. Notably, the digital models matched the Swi and Sor values reported in the 1968 study. It is noted that Sample E could not be compared to the previous report. This is highlighting a significant advantage of DSCAL in that it is not bound by lab experimental limitations such as inability to run SCAL programs on tight rock samples with permeability less than 1 mD.

Sample A

Sample B

Sample C

Sample D (Normalized)

Sample E

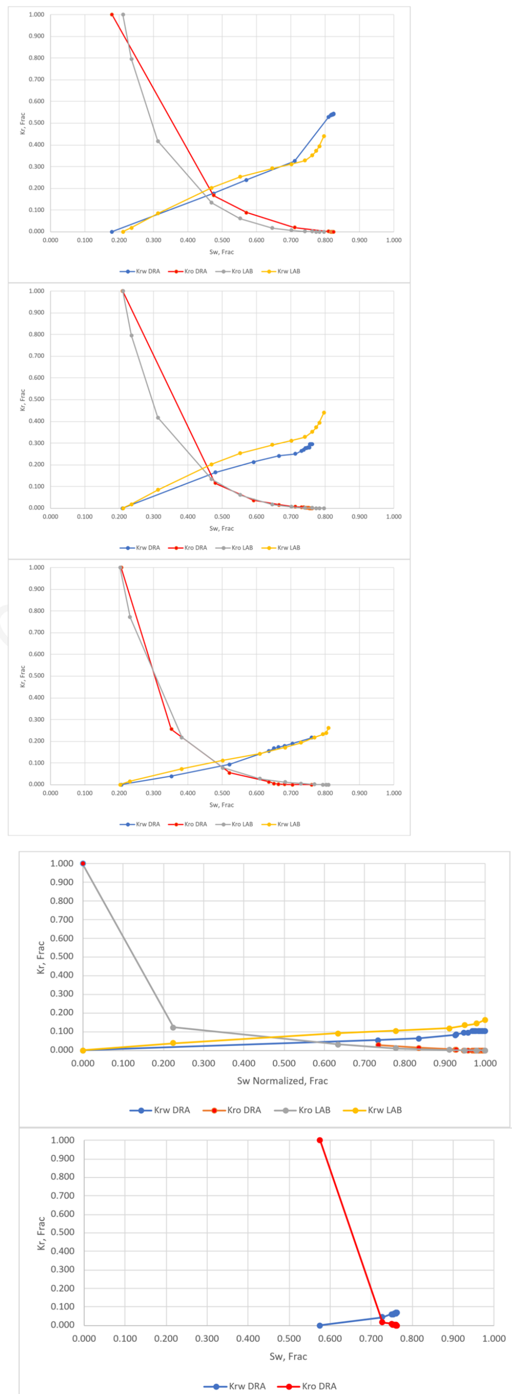

Figure 7—Digital and laboratory relative permeability curves as a function of water saturation for studied samples (digital SCAL data is shown with blue and red, laboratory data is shown using grey and yellow colors). Legend on the right provide sample IDs.

During the comparison of the relative permeability curves, 3 samples were found in the lab report that approximately matched the 5 plugs selected from the new sample set to cover the various flow units within Ratawi. Below is a comparison of the relative permeability curves. Sample C shows the closest overall match to the data set. For Sample D it is noted that Swi has high deviation compared to the lab data – however upon further investigation we found that the lab Swi does not align with the MICP data and the Swi in the report is likely in error and that the obtained Swi does better align with the DRA result.

Oil and water 3D saturations at Swi and Sor conditions for all studied samples are shown in Fig. 8. In 3D phase saturation figures, water is shown with blue color, oil is shown with red color, and rock is transparent. The flow occurs from left to right within individual 3D models.

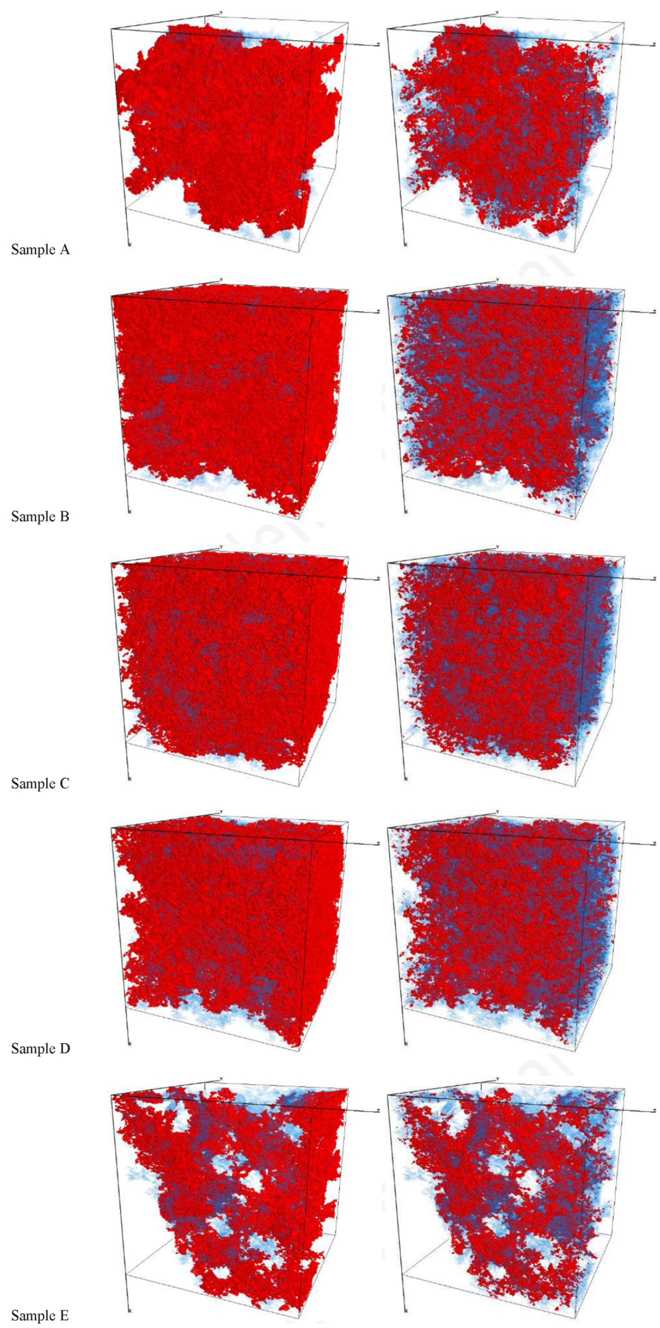

Figure 8—Oil (red) and water (blue) saturation at Swi (left) and at Sor (right) for all studied samples. Legend on the right provide sample IDs

While the USS modeling data was used in validation process with the historical data, the steady-state core flooding simulations with reservoir fluids were helped to obtain more accurate and representative results for today's dynamic reservoir models in the full saturation range. The digital steady-state (SS) oil-water imbibition relative permeability tests were performed. Oil and water were injected into the sample residing at irreducible water saturation Swi. SS digital experiments cover six or more injection ratios for imbibition each calculated with the DHD simulator to establish the associated relative permeability points. At each ratio, the injection continued until a two-phase pseudo-equilibrium was established, which is defined as a stabilization of phase saturations, phase production ratios and phase velocities. Once the equilibrium was reached for the current ratio, the digital experiment continued with the next injection ratio and at the final step only water was injected. The permeability to each phase was determined and recorded as a function of phase saturation and presented in charts. Phase permeability was normalized using oil permeability at irreducible water saturation (Swi).

These updated curves are recommended for use in reservoir simulation and development planning. An example of the steady-state data at current laboratory conditions is shown in Fig. 9 for Samples A and B. As one can see, there is a significant shift in crossover point when comparing legacy methods to todays methodology.

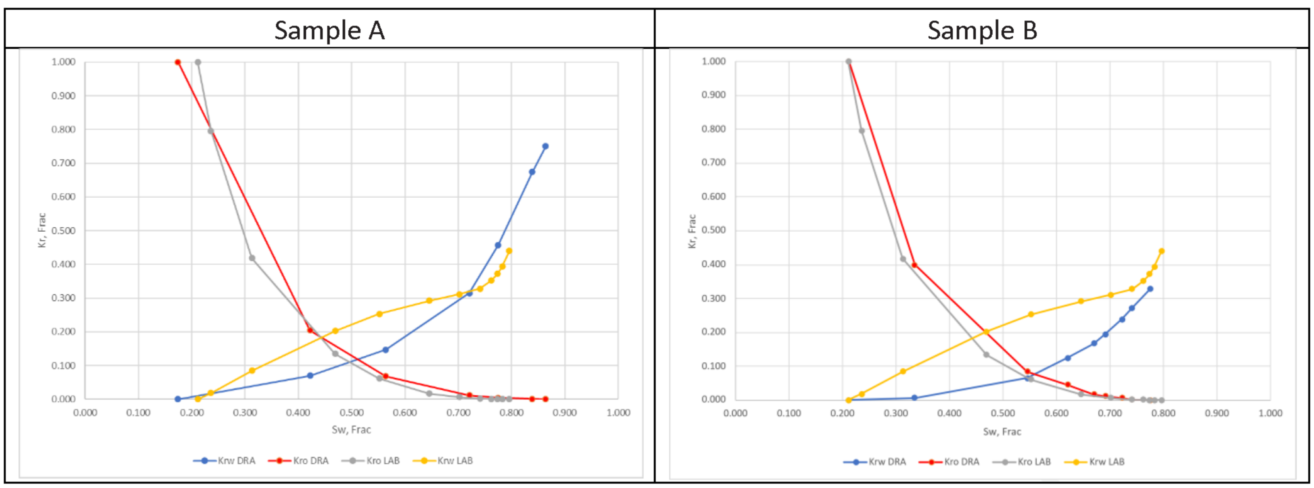

Figure 9—Digital steady-state relative permeability curves as a function of water saturation for sample A (left) and B (right) (digital SCAL data is shown with blue and red, laboratory data is shown using grey and yellow colors).

It is noted then from the results that

- Digital porosity and permeability values aligned fairly with RCA measurements.
- Pore throat distributions matched MICP data, confirming digital rock model integrity.
- The relative permeability curves demonstrated a generally favorable alignment with the legacy SCAL data; however, in instances where discrepancies occurred the DRA data better aligned than the 1968 data with other measurements from the 1968 data set such as MICP.
- DRA was able to model USS Relative permeability of samples that were not previously analyzed due to laboratory equipment limitations.
- Updated digital steady-state curves offer improved accuracy and are aligned with modern SCAL standards.

Conclusion

This study demonstrates the successful application of DRA and Digital SCAL workflows to characterize relative permeability in carbonate reservoir samples from the Ratawi formation in the Al-Khafji field. By integrating CT, micro-CT and SEM imaging techniques, mineralogy data, and applying DHD for pore-scale fluid simulations, the digital approach provided acceptable alternative to traditional SCAL methods. This is particularly valuable for samples that could not be physically tested due to equipment limitations or rock integrity concerns.

The digital rock models showed good agreement with RCA data, with porosity and permeability values generally within the same trend of laboratory measurements. Pore throat distributions derived matched well with MICP data, confirming the representativeness and resolution of the pore throat distribution.

After the confidence was established using RCA and MICP data, relative permeability curves generated under reservoir conditions aligned closely with historical SCAL data from the 1968 study, especially after accounting for methodological differences such as the use of non-reservoir fluids and legacy unsteady-state flooding data processing workflows. In cases where discrepancies were observed further investigation revealed that the digital results were more consistent with other experimental data, such as MICP on those samples in 1968, suggesting that the legacy measurements may have been affected by procedural limitations. This highlights the value of DSCAL not only in replicating historical data but also in refining and correcting it. It brings up an important point that while physical lab data is considered the "gold standard" and we use it to validate new technologies, it itself is not without potential errors. Lab measurements are also prone to issues that are not always flagged by the laboratory or the end consumer of the data as they are not aware of them.

Importantly, in this data set DSCAL enabled the modeling of samples that were previously excluded from SCAL due to tight permeability in Sample E. This capability extends the reach of core analysis programs and allows for more comprehensive reservoir characterization.

The updated relative permeability curves generated using reservoir fluids and simulating modern techniques also offer improved accuracy and alignment with contemporary SCAL standards. This updated data can support better long term field development and SCAL data influences strategies when dealing with aged assets such as application of Enhanced Oil Recovery solutions.

In summary, the study confirms that:

- Digital SCAL workflows can fairly replicate and enhance traditional SCAL results.
- DSCAL is a viable solution for re-characterizing legacy datasets and overcoming limitations of historical laboratory methods.
- The integration of imaging, mineralogy, and fluid simulation offers a scalable and repeatable approach for future reservoir studies.

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